Shared Stopping and Necessary Temporal Contracts: A Native Mechanical Component Study

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Abstract

A recovery that stops one process on time can prematurely terminate another required operation. We test this conflict in 432 native two-rotor MuJoCo control rollouts, with selective commands sharing a serial channel and a global relay stopping both processes. Counting only process A substantially overstates joint outcomes. A deterministic joint-contract scheduler yields 22/108 joint timing TRUE outcomes, versus 18/108 for an A-only filter and 20/108 for ordinary retry. Its five repairs and three harms relative to retry expose a tradeoff between conservative failure support and successful opportunistic recovery. These are mechanical development results, not learned robot or chemical-success evidence.

1. Multiple necessary conditions

A prior single-rotor component favored a reliable, cost-free early stop. We now test an explicit competing requirement. Each process has a measured reference event τ_i and a physical completion T_i. A requires $T_A - \tau_A \in [0, U_A]$; B requires $T_B - \tau_B \in [L_B, U_B]$ and no low-speed interruption before its minimum duration. A global stop can satisfy A while violating B. The joint temporal verdict is FALSE if any necessary condition is FALSE, TRUE only if all are TRUE, and UNKNOWN otherwise. Later completion does not erase an earlier interruption of the original attempt.

For a finite recovery branch b, a conditional candidate must respect every required operation, not just the failed one:

$$\begin{aligned} &\forall b \in \mathcal{B}(a), \quad i \in \{A, B\} : \\ &[T_{i,b}^{-} - \tau_i^{+}, T_{i,b}^{+} - \tau_i^{-}] \subseteq [L_i, U_i]. \end{aligned} \tag{1}$$

This is an elementary interval check under execution and duration support, not a new temporal guarantee. Our finite graph includes retry success, retry failure, feedback wait and a fixture-reliable relay. The joint control also evaluates waiting until a relay's B completion supports its minimum duration; that wait simultaneously consumes A's deadline.

2. Frozen development experiment

Two contact-free rotors use 10 ms integration, nonnegative drive and a latched shared stop interface; electronics, fluids and chemistry are not modeled. Each trajectory's reference is its own first shaft-angle crossing of 2 rad. Completion starts a speed ≤ 0.1 rad/s segment lasting 0.2 s. Low-speed samples preceding B's minimum witness an interruption; ambiguous boundaries remain UNKNOWN.

The 108 cases cross three A responses, three delivery delays, two A deadlines, two B windows and three A-fault conditions; B's response is publicly calibrated. Four equally privileged controls compare global stopping after a delivered trigger, selective retry, A-only filtering and deterministic joint scheduling. Selective requests share a non-preemptive channel. ACK indicates receipt; delivered power feedback reveals a dropped action.

Actors use delivered encoder/power readings, public command schedules, windows and fixed development models; true response/fault labels are withheld. There are 72 calibrated and 36 shifted cases. Response variation is not unseen task-graph generalization. Controls react after delivered references within the finite graph, not all anticipatory or MPC strategies.

We freeze source/model/calibration/dependency identities after 11 tests and eight diagnostics. The main 432 rollouts contain 432,432 samples and 4,752 recorded RGB frames, excluding diagnostics. Controls do not read RGB. A separate array event oracle recomputes all scores; native reconstruction including fault application has zero discrepancy. All 108 groups share pre-ACK mechanical prefixes. Public-history replay reproduces every decision and queue, reusing the frozen control as a causal check.

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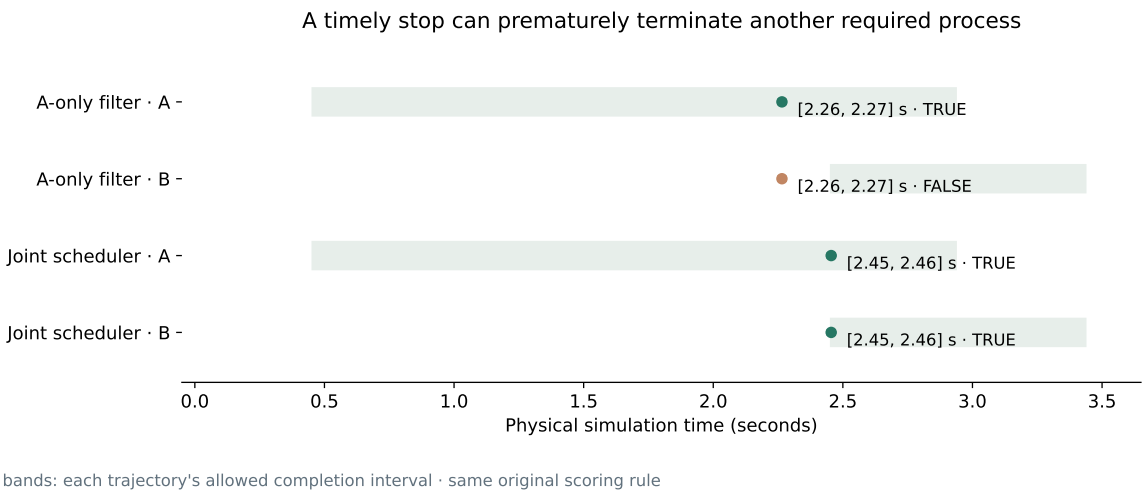


Figure 1. One measured persistent-A-fault case within calibration. A's window is $[0, 2.5]$ s and B's is $[2, 3]$ s after their respective references. The A-only filter stops both at $[2.26, 2.27]$ s: A timely, B premature, joint FALSE. The joint scheduler stops at $[2.45, 2.46]$ s, meeting both windows and B continuity. Shading uses each measured reference; no score or window is rewritten after execution.

Table 1. All 108 parameter cases per control. A TRUE alone conceals violations of another necessary operation. T/F/U are joint verdicts.

Control	A TRUE	T	F	U
Global relay	42	3	102	3
Selective retry	21	20	85	3
A-only filter	34	18	88	2
Joint scheduler	22	22	84	2

3. Gains, harms and remaining failures

Only three of the global relay's 42 A-timely cases are jointly TRUE. B has 90 definite interruptions under global stopping, 36 under A-only filtering, six under ordinary retry, and none under joint scheduling. Preserving B also lowers A-timely counts relative to the A-only filter. The joint control therefore does not improve every metric.

In the calibrated subset, joint scheduling produces 20 TRUE, 50 FALSE and two UNKNOWN outcomes. Under response shift it yields two TRUE and 34 FALSE, the same TRUE count as retry and A-only filtering. Six selected predictions with both timing supports eligible nevertheless have a non-TRUE actual conjunction: four joint-scheduler and two A-only trajectories, all outside calibration and representing four parameter cases. No such in-range counterexample is observed; finite samples do not establish coverage.

Relative to A-only filtering, the joint control repairs four cases and harms none. Relative to global stopping it repairs 21 but harms two. Relative to ordinary retry it repairs five and harms three, for a net gain of two. The three latter harms are recoverable first-drop faults with B's three- second minimum: requiring the persistent-failure continuation rejects a retry that actually could succeed. Protecting B with a later relay then misses A's deadline. Conservative branch admission is therefore not equivalent to maximizing empirical success. We retain every harm, UNKNOWN, infeasible setting and repeated baseline configuration.

4. Research implications and scope

Recovery must account for effects on other necessary operations. Deterministic joint control explains this gain, without evidence of agentic reasoning or weight absorption. Next comparisons require public response identification/calibration and matched successful-retry versus collateral-risk treatment. Subsequent history-conditioned policy and student-state labeling comparisons must match teacher budgets and separate context/parameter updates from runtime restrictions.

Learned-policy runs, real robot trials, chemical successes and independent task/batch clusters are zero. Rotor actions are not admitted as robot BC or chemical RL data. Logged host work does not advance simulation; declared transport/actuator delays do. Real-time latency, inferential significance, global safety and complete Paper B acceptance remain unvalidated.